

Course Glossary - Generative AI for Solutions Architects and Systems Analysts

Welcome! This alphabetized glossary contains many terms used in this course. Understanding these terms is essential when working in the industry, participating in user groups, and participating in other certificate programs.

Term	Definition
Acceptance criteria	A set of clear conditions that a system or feature must meet for a user story to be considered complete and acceptable.
AI-driven communication plan	A structured plan automatically generated by generative AI that defines stakeholders, key messages, communication channels, and timing for project updates.
AI-generated artifact	A professional document, template, or diagram created using generative AI to accelerate project documentation and ensure consistency.
AI-powered validation report	A summary automatically produced by AI that reviews test results, identifies patterns in failures, and suggests corrective actions.
Application programming interface (API)	A defined set of rules and connections that allow software applications to communicate and share data with each other.
Assumptions	Statements taken to be true without direct evidence during requirements or system design, used as the basis for planning or analysis.
Backend	The part of a software system that handles data processing, storage, and server-side operations hidden from the user interface.
Baseline requirement document	A finalized version of a requirements document that serves as a consistent reference for analysis, design, and testing.
Benchmarking	The process of comparing system or process performance against standards or best practices to identify areas for improvement.
Business process model and notation (BPMN)	A standardized graphical method for visualizing business processes using symbols to represent tasks, decisions, and flows.
Checklist	A structured list of items or actions used to verify that all steps in a process, test, or deployment are completed correctly.
Communication template	A reusable framework created with AI that structures project messages such as status updates, stakeholder emails, or meeting minutes.
Constraint	A limitation or restriction on project scope, resources, or system performance that must be considered during planning.
Customer journey map	A visual diagram that shows the stages a customer experiences when interacting with a product or service, including actions, emotions, and pain points.
Data flow diagram (DFD)	A visual representation showing how data moves between inputs, processes, and outputs within a system.
Deployment checklist	A structured list of environment-specific steps, created by AI, that ensures a system is deployed safely and completely.
Deployment environment	The setting where software runs, such as development, staging, or production, each used for different testing or live operations.
Edge case	A scenario that occurs under extreme or unusual conditions, often tested to ensure the system handles unexpected inputs.

Executive summary	A brief, high-level overview created with AI that highlights the main actions, decisions, and outcomes from a report or meeting.
Functional requirement	A description of what a system should do, focusing on features, behaviors, and interactions visible to users.
Generative artificial intelligence (generative AI)	A branch of artificial intelligence that can produce new text, images, or data patterns by learning from existing information.
Given / when / then format	A structured format for defining test cases that describe initial conditions (given), actions (when), and expected outcomes (then).
Integration point	A connection where two systems, components, or processes exchange data or interact.
Iterative prompt refinement	The process of repeatedly improving an AI prompt by adding clarity, context, or constraints until the output meets requirements.
Load testing	The process of evaluating how a system performs under high user demand or stress conditions, often generated by AI simulations.
Message flow	In BPMN, the line that shows the communication or exchange of information between different participants or systems in a process.
Multilingual collaboration	The practice of using AI tools to translate, summarize, or adapt communication for teams working across multiple languages and cultures.
Negative test case	A scenario intentionally designed to ensure a system properly handles invalid, unexpected, or incorrect inputs.
Non-functional requirement	A description of system attributes such as performance, security, usability, or reliability that define quality rather than behavior.
Performance testing	The practice of assessing how quickly and efficiently a system performs under expected workloads.
Process modeling	The activity of visually representing business or system processes to analyze, improve, or automate workflows.
Prompt engineering	The craft of writing clear, specific, and structured instructions that guide generative AI to produce accurate and useful outputs.
Requirement gathering	The process of collecting stakeholder needs and expectations to define what a system must accomplish.
Requirement validation	The activity of ensuring that documented requirements accurately reflect stakeholder intentions and are feasible to implement.
Rollback plan	A predefined strategy for restoring a system to a previous stable state in case a deployment fails.
Service task	In BPMN, an automated activity performed by a system or service without direct human involvement.
Simulation scenario	An AI-generated role-play or case setup that imitates stakeholder interviews or system behavior for training or testing purposes.
Stakeholder	Any individual or group affected by, involved in, or able to influence a project's outcomes.
Stakeholder interview template	A structured set of questions designed to gather requirements and insights from different stakeholder groups.
Stakeholder map	A diagram or table generated by AI that identifies key stakeholders, their influence, interests, and preferred communication methods.
Stress test scenario	A test designed to determine system behavior under extreme workloads, network issues, or failure conditions.
Structured prompt	A formatted AI input that includes role, task, context, and output format to ensure clear, targeted responses.
System architecture diagram	A visual representation showing the components, data flows, and interactions within an entire system.

System deployment	The process of releasing a software system into a live environment for real use by end users.
Test case	A detailed, step-by-step set of conditions and actions used to verify whether a system meets its specified requirements.
Test script	An automated program generated by AI that executes test cases and validates software functionality.
Trigger word	A word or phrase such as “if,” “when,” or “after” that signals process steps, decisions, or conditions within unstructured text.
Use case	A detailed description of how a user interacts with a system to achieve a goal, often including actors, flows, and alternate scenarios.
User story	A short, structured statement that describes a desired feature from an end-user perspective, typically following the format “As a [role], I want [goal], so that [reason].”
Validation report	A structured document, often AI-generated, summarizing test outcomes, highlighting errors, and recommending next steps.
Workflow optimization	The practice of using AI tools to review, refine, and improve the efficiency of existing business processes.